Roll No.: ……………..………………

END SEMESTER EXAMINATION NOVEMBER 2025

Name of the Program: B.Tech in Petroleum Engineering Semester: V

Name of the Course: Machine Learning for Reservoir Characterization Course Code: TPE-511

Time: 3 Hours **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**
ii. Answer any two sub questions among a, b & c in each main questions
iii. Total marks in each main question are twenty.
iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	What is the fundamental goal of Exploratory Data Analysis (EDA), and how do its two primary components—Visualization and Descriptive Statistics—work together to achieve this goal?	CO1
(b)	Describe the appropriate use case for a Histogram versus a Box Plot during the EDA phase. Specifically, what different aspects of a single numerical variable's distribution does each visualization technique effectively reveal?	CO1
(c)	Identify and define the three main categories of Descriptive Statistics: Measures of Central Tendency, Measures of Dispersion (or Variability), and Measures of Position. Provide one example statistic for each category.	CO1
Q2	(20Marks)	
(a)	The following table contains historical data for 8 customers, used to predict whether they will Buy a Laptop.	CO2

ID	Credit Rating	Income	Student	Buys Laptop (Target)
1	Fair	High	No	No
2	Excellent	High	No	No
3	Fair	High	Yes	Yes
4	Excellent	Medium	No	Yes
5	Fair	Low	Yes	No
6	Excellent	Low	Yes	Yes
7	Fair	Medium	No	Yes
8	Fair	Medium	Yes	Yes

- Calculate the Initial Entropy (H(S)) of the full dataset based on the target variable 'Buys Laptop'.
- Calculate the Information Gain for the features Income and Student.
- Identify the Best Feature for the root node split between these two features.

(b)	Both Linear Regression and Logistic Regression are fundamental supervised learning algorithms. State the primary difference between the type of problem each algorithm is designed to solve and the function they use to map the input to the output. The following table shows data relating a student's study hours (x_i) to their test score (y_i).	CO2

Observation (i)	Study Hours (x_i)	Test Score (y_i)
1	2	5
2	3	7
3	5	10
4	7	11
5	8	13

Calculate Test Score (y_i) for Study Hours (x_i) = 4 with Linear Regression.

(c) When evaluating a classification model, explain the difference between Precision and Recall. Provide a scenario where maximizing Recall would be more critical than maximizing Precision.

Q3 (20Marks)

(a) The following table shows data for 6 patients, including their weekly hours of moderate exercise (x) and whether they developed a specific health risk (Y). CO4

Patient ID	Exercise Hours (x_i)	Health Risk (Y_i)
1	1	1 (Yes)
2	2	1 (Yes)
3	4	1 (Yes)
4	5	0 (No)
5	7	0 (No)
6	8	0 (No)

CO3
CO3

Assuming Exercise Hours (x_i) = 4.5 calculate the Health Risk (Y_i) with Logistic Regression.

(b) Reservoir datasets often contain both static and time-series components. Explain how the processing objectives and mathematical techniques used for static data fundamentally differ from those used for time-series signal processing.

(c) What unique data structures or indexing requirements must a Spatio-Temporal Database satisfy that a traditional relational database (SQL) does not, specifically for efficient querying and analysis of reservoir data (e.g., tracking a pressure front moving through a 3D grid over time)?

Q4 (20Marks)

(a) Generalized Linear Models (GLMs) extend Ordinary Least Squares (OLS) Linear Regression. Identify the two main structural components that differentiate a GLM from a standard OLS model, and explain how these components allow GLMs to model non-Normal response variables. CO5 CO5 CO4

(b) In unsupervised learning, internal evaluation metrics are used to assess clustering quality. Explain the calculation and interpretation of the Silhouette Coefficient. What does a Silhouette Coefficient close to +1, 0, and -1 imply about a data point and its assigned cluster?

(c) Beyond simply tracking raw metrics, Advanced KPI Identification requires linking analytical outputs to strategic value. Explain the critical difference between a Lagging Indicator and a Leading Indicator KPI. Provide an example of each type of indicator that could be derived from a predictive model's output.

Q5 (20Marks)

(a) Distinguish between Total Porosity (ϕ_T) and Effective Porosity (ϕ_E). Explain which of these two porosity values is the primary controlling factor for fluid flow and why, relating your answer to the connectivity of the pore network. CO5

CO6

(b)	Define Capillary Pressure (P_c) in the context of reservoir rock properties. Explain how the concepts of Wettability and Capillary Pressure curves are used together to estimate hydrocarbon saturation and determine the height of the Transition Zone within a reservoir.	CO6
(c)	Define the Skin Factor (S) determined from a well test. In the context of building a Machine Learning model (e.g., a Regression model) to predict initial production rates across a field, explain how and why the calculated Skin Factor is a crucial feature to include in your model's input dataset.	